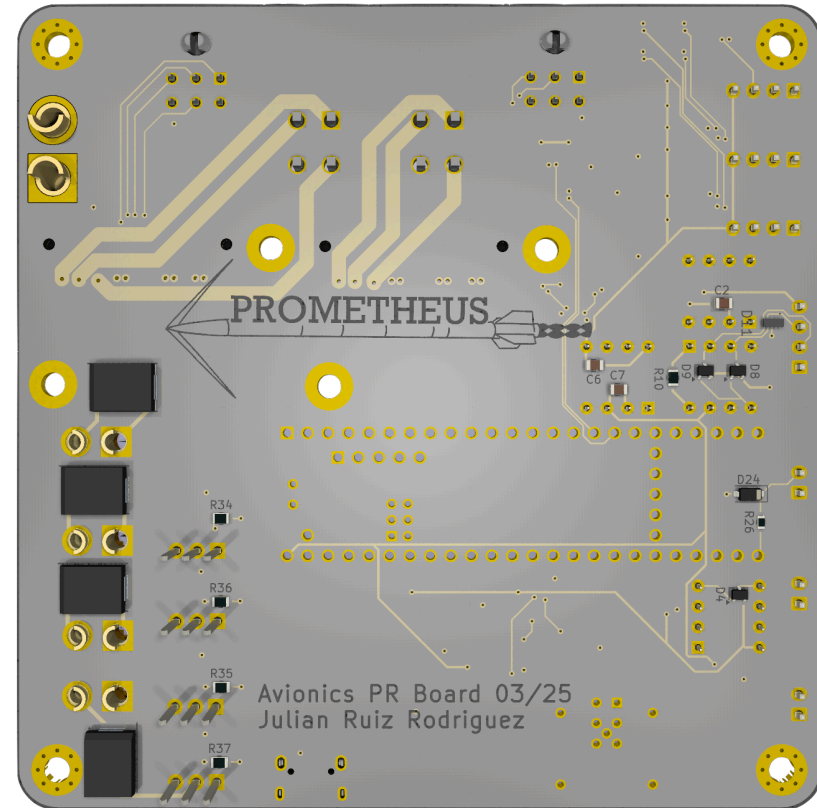
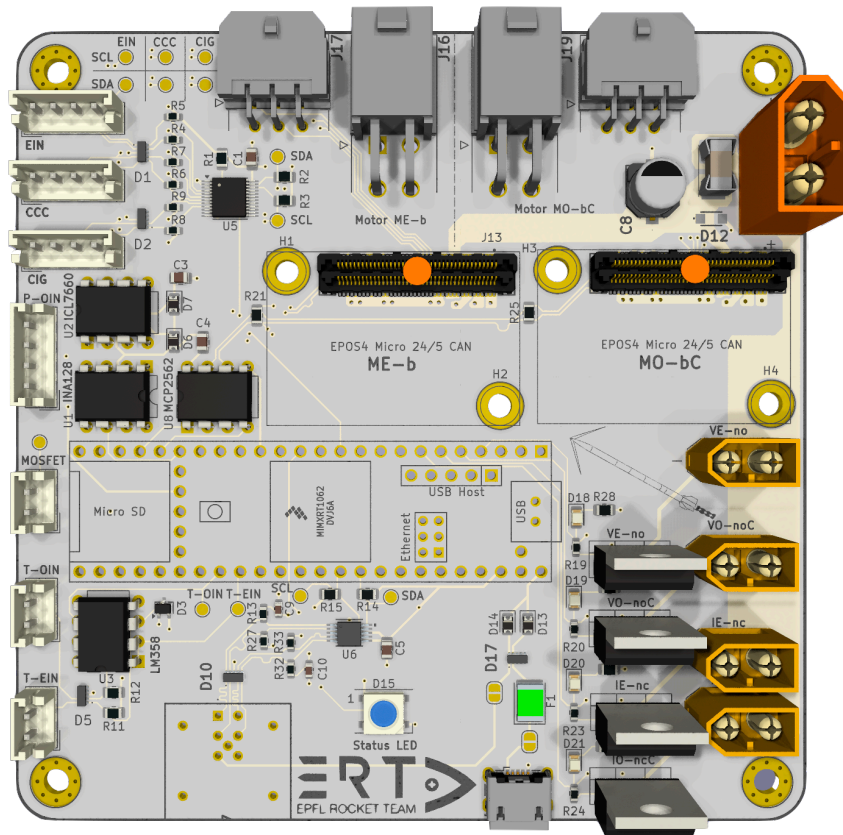


PROMETHEUS

Propulsion Regulation for Optimal Motor Engine Thrust, High-accuracy Electronic Utility System



DESIGN NOTES:
Example text for
informational
design notes.

DESIGN NOTES:
Example text for
cautionary design
notes.

DESIGN NOTES:
Example text for
critical design
notes.

LAYOUT NOTES:
Example text for
critical layout
guidelines.

DEBUG NOTES:
Example text for
debug notes.



PROMETHEUS

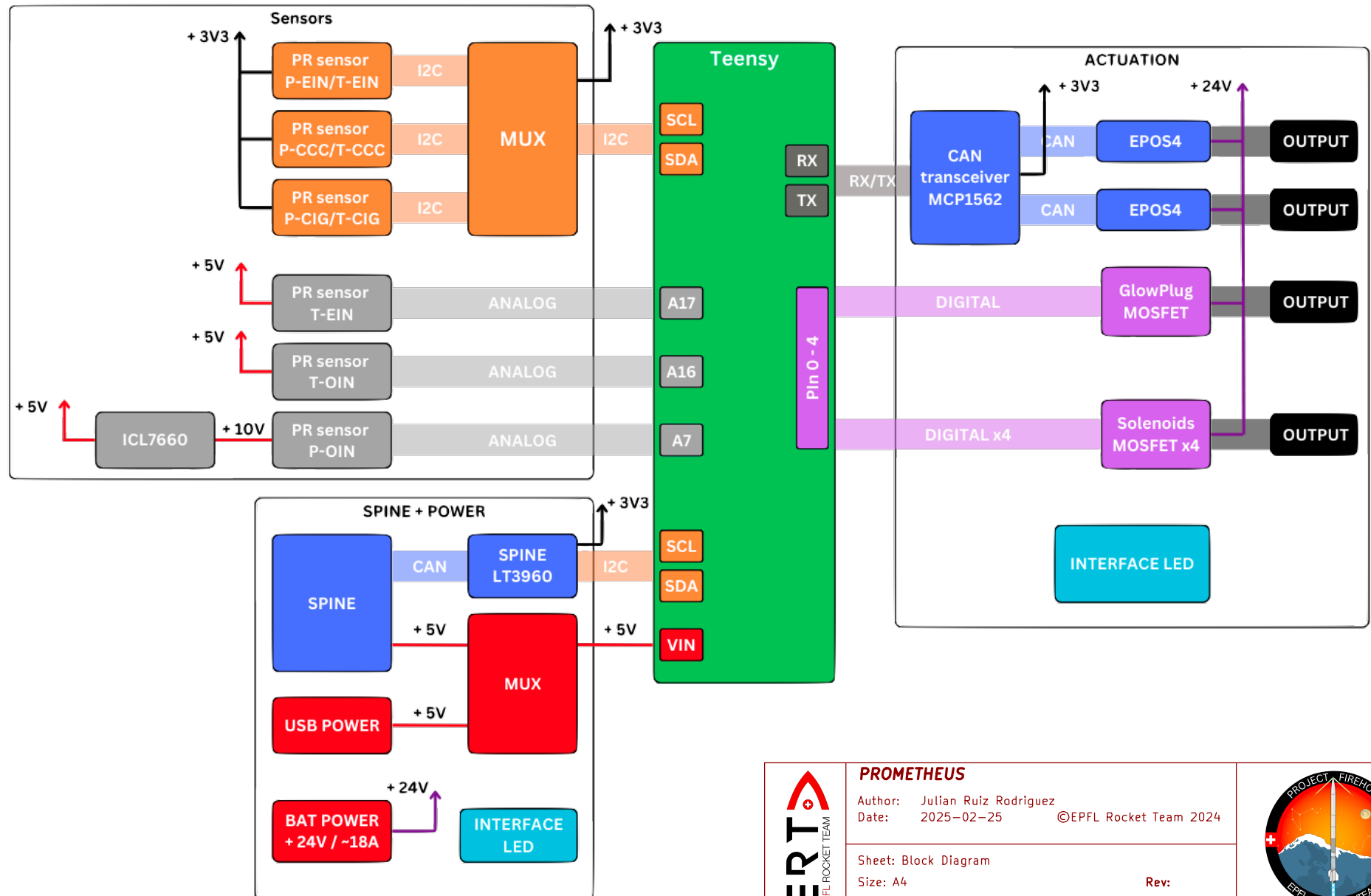
Author: Julian Ruiz Rodriguez
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Sheet: Root
Size: A4
Software: KiCad E.D.A. 9.0.3

Rev:
Id: 1/9



Block Diagram



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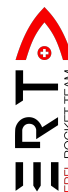
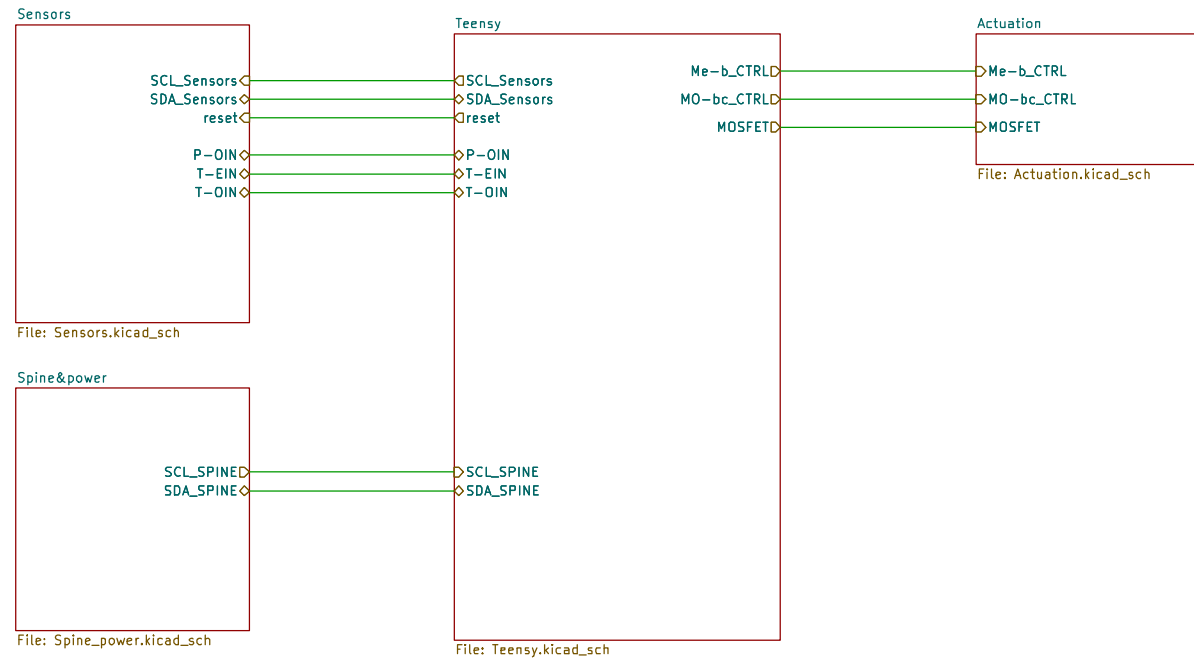
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Sheet: Block Diagram
Size: A4
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Id: 2/9



Project Architecture



PROMETHEUS

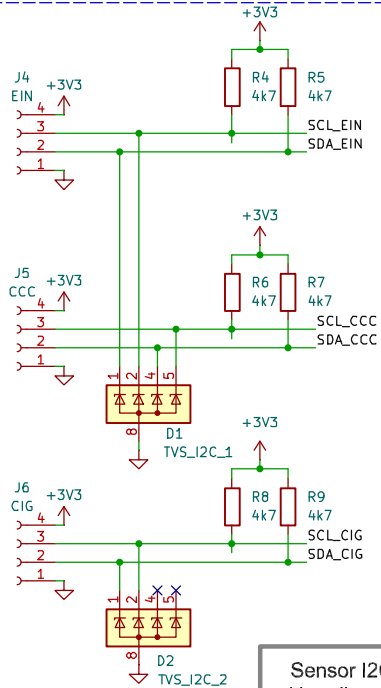
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Sheet: Project Architecture
Size: A4
Software: KiCad E.D.A. 9.0.3

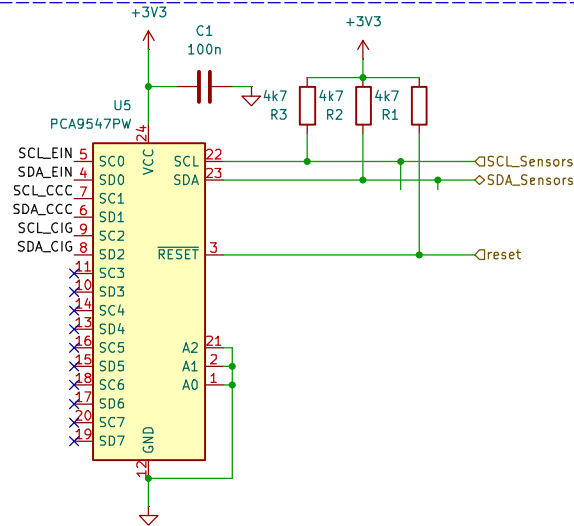
Rev:
Id: 3/9



EIN / CCC / CIG (DIGITAL I2C)

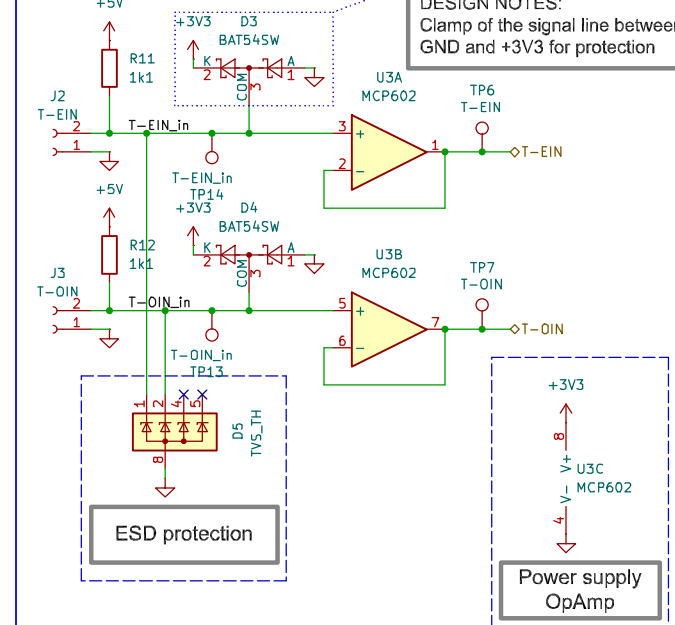


Sensor I2C input
with pull up resistors



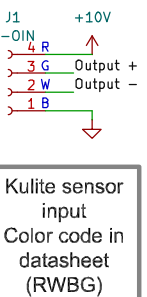
DESIGN NOTES:
I2C MUX for 3 sensors
Need pull up resistors before and after the MUX

T-EIN / T-OIN (ANALOG)

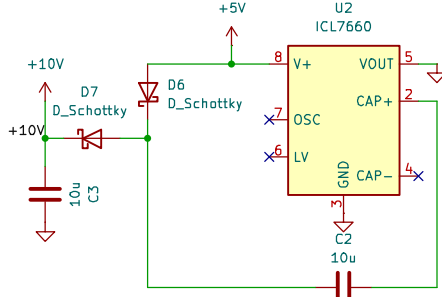


DESIGN NOTES:
Clamp of the signal line between
GND and +3V3 for protection

Design of the resistor and ampli op :
J2 & J3 are input from a thermistor with value between 1k0 and ~1k7 Ohms
Tension divider output with $R = 1k1 \rightarrow$ between 2.39V and 3.04V
Using +5V to reduce load on the Teensy



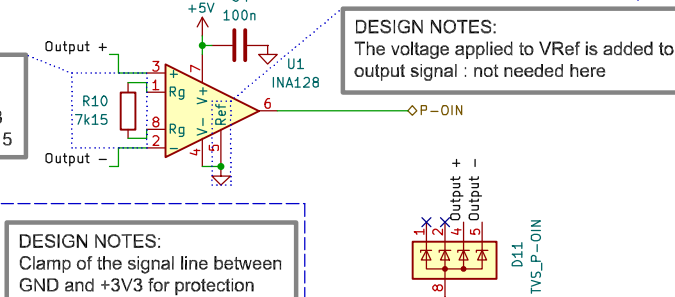
Kulite sensor
input
Color code in
datasheet
(RWBG)



DESIGN NOTES:
Positive voltage doubling to
achieve the electrical excitation
of the Kulite sensor (+10 VDC)

DESIGN NOTES:
we have $V_{out} = 2 \cdot V_{in} - 2 \cdot V_f$
with V_f the voltage drop at
the Schottky diodes

DESIGN NOTES:
Full Scale Output : 100mV $\rightarrow$
 $G=33$ ampmplification up to 3V3
 $G = 5 + 200k/R10 \rightarrow R10=7k15$



DESIGN NOTES:
The voltage applied to V_{Ref} is added to
output signal : not needed here

DESIGN NOTES:
Clamp of the signal line between
GND and +3V3 for protection



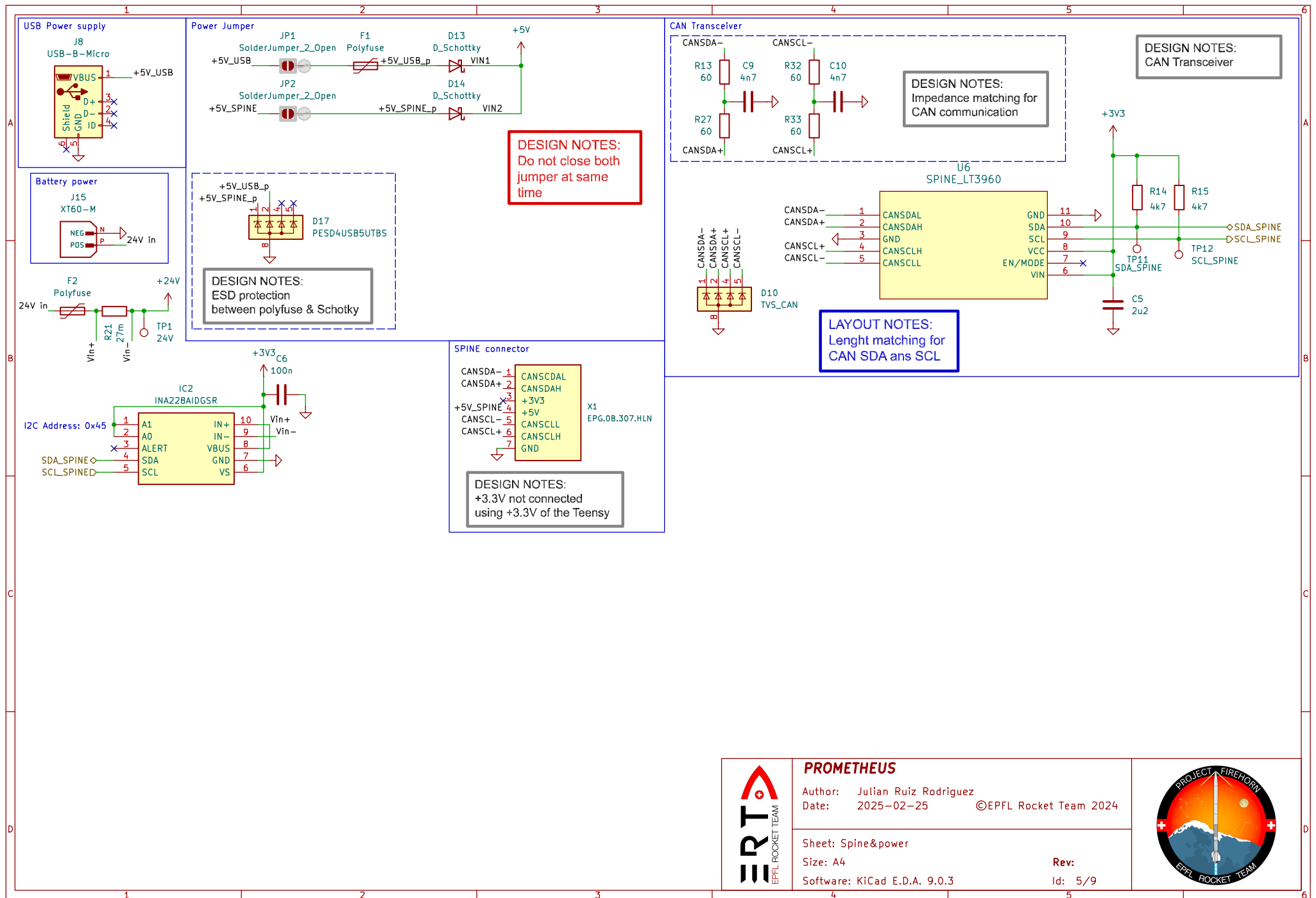
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Sheet: Sensors
Size: A4
Software: KiCad E.D.A. 9.0.3

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Sheet: Spine&power

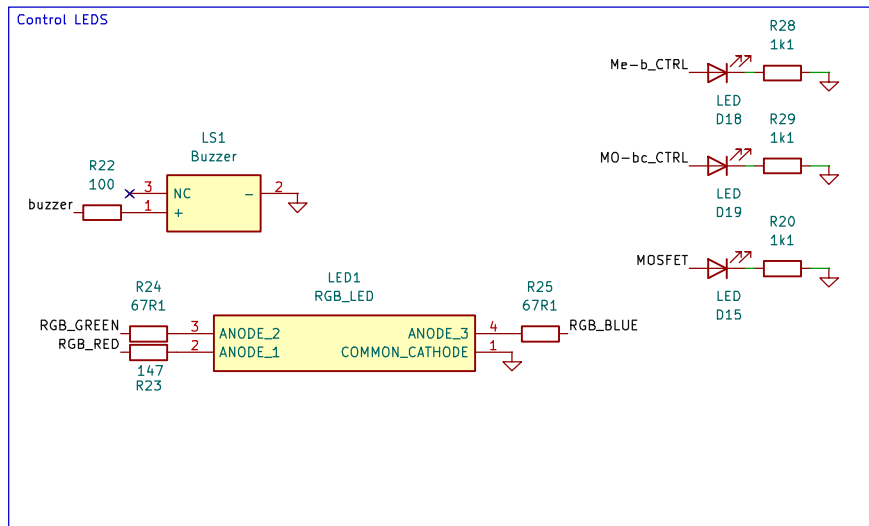
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Rev:

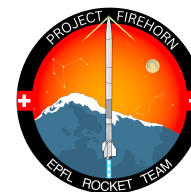
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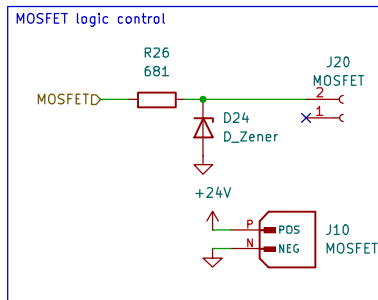
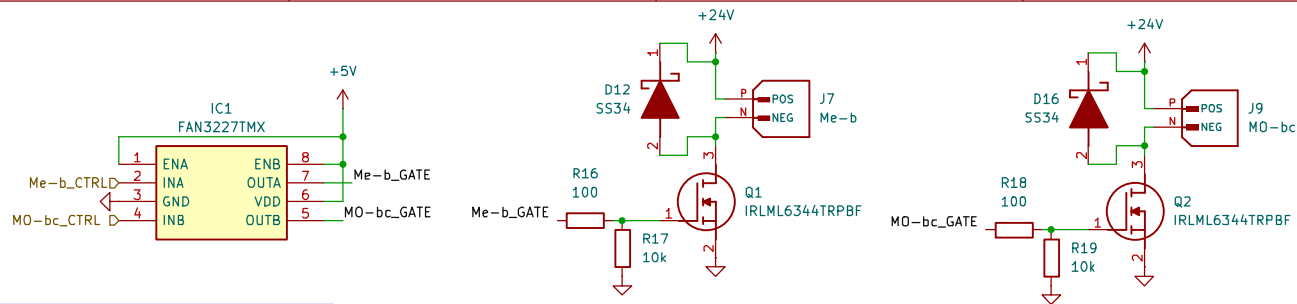




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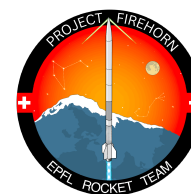
Sheet: Actuation

Size: A4

Software: KiCad E.D.A. 9.0.3

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NAME	SOURCE	LEVEL
+3V3	Teensy	??V ± ??%
+5V	SPINE	??V ± ??%
+5V	USB POWER	??V ± ??%
+24V	BATTERY	??V ± ??%

Power Sequencing Diagram of your PCB here

DD.MM.YYYY - xxx Revision
Variant: xxx

[illegible]

DD.MM.YYYY - xxx Revision
Variant: xxx

[illegible]

DD.MM.YYYY - xxx Revision
Variant: xxx

[illegible]DD.MM.YYYY - xxx Revision
Variant: xxx[illegible]

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Sheet: Revision History

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Software: KiCad E.D.A. 9.0.3

Rev:

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